SPEAR3 LLRF Upgrade

Interface Chassis and Waveform Buffer Design Requirements

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This document sets the design requirements of the Interface Chassis and the Waveform Buffer, the two boards of one chassis in the SPEAR3 LLRF Upgrade: every input and output with its medium, level and sense; the logic that forms the permits, written from the required behavior of the system; latching, reset and first-fault recording; timing and fail-safe states; the Waveform Buffer channels, buffers, comparators and collector-power trip; status reporting to the RF MPS and EPICS; and verification. It covers the transition period, when the chassis replaces the Fast Interlock Chassis and the RF Local Display Panel while the legacy station is still in service, and the end state. Statements that still need confirmation are tagged **[O*n*]** and collected in the Open Items register; values marked *proposed* are for review.

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Revision History

Revision	Date	Scope of Change	Summary
R0	2026-11-10	Draft for review	Initial issue: interfaces, signal levels, permit logic written from the required behavior of the system, latching, reset and first fault, timing and fail-safe states, Waveform Buffer channels, buffers, comparators and collector-power trip, status reporting and verification, for the transition period and the end state. Every signal checked against the drawings of the panel, the Fast Interlock Chassis and the AIM: the X530 lines that the panel only displays are not taken over, nor are the key switches, which operators do not use, or the beam-abort output, which nothing at SPEAR3 receives.

Open Items — Still To Be Confirmed

Each item is tagged **[O*n*]** where a requirement depends on it; the page column links to that point. Where an item is the same as one of Doc 0 or Doc M, that item is named.

ID	Item	What is needed	Needed before	Page
O1	Logic equations	The equations of §5 are written from our understanding of the system's required behavior, not recovered from the panel's GAL22V10. Review them with J. Sebek and the HVPS and SPEAR MPS owners, and compare them with the recorded behavior of the panel (baseline I0-T1 of Doc M); keep or remove each difference deliberately (Doc M O1).	Logic design freeze (I2)	p. 17
O2	LLRF9 input option	The factory-set logic level of the interlock inputs of the installed LLRF9 units: 3.3 V, 5 V or 24 V (Doc 0 O1).	Board design	p. 10
O3	ARC4 interface	The lines of the two system modules and the test module: their number, levels, the form of the fired-detector identity (parallel bits or an encoded word), the latch reset and the test lines (Doc 0 O5).	Board layout	p. 12
O4	Legacy AIM and Fast IC signals	Confirm on the hardware the J15 levels and sense that the drawings give (Table 15) and measure the length of the reset and timeout-bypass pulses; the RF level at the detector input J32 and the legacy detector's transfer curve, on the bench or from archived readings; what drives the Fast Interlock Chassis's spare-fault input, and the AIM's arc-channel and spare-fault enables (Doc M O3).	Production design	p. 14
O5	X530 pins	The pin and terminal numbers of the J2 and J3 lines, on which the drawings disagree, checked by continuity (Doc M O2).	Cable fabrication	p. 14

ID	Item	What is needed	Needed before	Page
O6	Heartbeat	The period and duty cycle of the PLC-5 heartbeat at the panel output and at the AIM input, which set the watchdog timeout (Doc M O4).	Watchdog design	p. 11
O7	SPEAR MPS and orbit permits	Their wiring into VXI Slot 5 and the current their sources can supply to a second input (Doc M O6).	Input design (I6)	p. 11
O8	Waveform Buffer channels	Channel 1 (klystron drive power) and channel 7 (klystron forward power for the collector power) are proposed; channel 8 is spare (Doc 0 O4).	RF hookup (I2)	p. 23
O9	HVPS inductor voltages	The pick-off points, amplitudes and cable route of the two inductor-voltage signals in B118.	Waveform Buffer HVPS inputs (I2)	p. 24

1 Purpose, Scope and Basis

Purpose. The Interface Chassis is the hardware interlock hub of the upgraded SPEAR3 RF station: it sums the permits of the subsystems and the external safety systems, drives the permits to the LLRF9 and the HVPS controller, tells the SPEAR MPS when the station is not ready for beam, latches every fault and records the first one [R1], [R4]. The Waveform Buffer, on the second board of the same chassis, digitizes the RF and HVPS signals that the LLRF9 does not see, keeps them in circular buffers that freeze on a fault, and trips through the Interface Chassis on its own comparators and on the klystron collector power [R1], [R5]. This document sets the design requirements of both: their functions, every input and output with its medium, level and sense, the permit logic, timing, fail-safe states, status reporting and verification. It is the reference for the production review of the chassis (Doc M A6) and for its bench and installation tests (Doc M I2-T1 to I2-T4) [R3].

Scope. Both boards, the chassis, its connectors and its EPICS interface. The chassis is installed at integration I2 of Doc M, in place of the Fast Interlock Chassis 340-308 and the RF Local Display Panel 340-310, while the AIM, the RFP, the PLC-5 and the legacy HVPS controllers are still in service; it keeps the same role through the end state after the VXI crate is retired at I8 [R3]. Out of scope: the ARC4, the LLRF9, the HVPS controller, the RF MPS program and the PPS Interface Box. The chassis carries no personnel-protection function.

Basis. The upgrade design, Doc 0 [R1]; the legacy station, Doc L [R2]; the integration plan, Doc M [R3]; J. Sebek's specification of the Interface Chassis and design notes on the Waveform Buffer, the power detectors, the analog components, the controller interfaces, the MPS requirements and the HVPS fibers [R4]–[R12]; the drawings of the panel, the Fast Interlock Chassis, the AIM and the RF MPS wiring [R13]–[R18], [R21]; the decoded RF MPS PLC-5 program [R19]; the LLRF9 manual [R20].

Conventions.

- A requirement carries an ID and uses "shall". IC marks the Interface Chassis board, LG its logic, WB the Waveform Buffer board, ST status reporting and EPICS, CH the chassis. Values marked *proposed* are design proposals for review.
- Every permit is active-high: 24 V present, a lit fiber or a high LLRF9 output gives the permit, so a broken cable, a dark fiber or a lost supply removes it [R1], [R8]. The light conventions of the legacy fibers are kept.

- A permit that is removed latches as a fault until a reset; a reset never clears a fault whose cause is still present.
- Integration numbers I2 to I8 and test numbers are those of Doc M [R3].

Terms.
Permit input An input that must be present for some output to give its permit.
Command input An input that requests an action, such as the HVPS-on request; it does not latch.
Act mask A hardware selection, one per input, that decides whether the input acts on the outputs. An input whose mask is off is still read, latched and reported (monitor-only).
Transition interface An interface of the Fast Interlock Chassis or the panel that the chassis must present while the legacy station is in service.
End-state interface An interface of J. Sebek's specification [R4], used once the subsystem behind it is integrated.

2 Role in the Station

Table 3 lists the functions that the chassis takes over from the legacy units and those that are new.

Table 3: Functions of the chassis.

From	Function	In the chassis
Fast Interlock Chassis	The six AIM commands on J15 turned into fibers to the panel; the AB summary and heartbeat returned to the AIM [R12]	AIM interface, §4.8
Fast Interlock Chassis	Klystron forward-power detector: RF input J32 and output J31 ("RF AMPL.") on the rear, cabled to the PLC-5 for its collector-power calculation; the front monitor output J16 is not cabled [R12], [R14], [R18]	Detector, §4.10
Fast Interlock Chassis	Arc-detection front end and its cable to the AIM's J4; test-light fibers; spare-fault input [R14]	Not taken over (Table 4)
RF Local Display Panel	The logic that forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort; two latches for the beam abort and the crowbar [R13]	Permit logic, §5
RF Local Display Panel	End of the three HVPS fibers [R10], [R11]	HVPS fibers, §4.1
RF Local Display Panel	The X530 lines that its logic and fibers use; buffering of the HVPS voltage and current for the PLC-5 [R13], [R16]	PLC-5 interface, §4.9
RF Local Display Panel	Beam-abort output on BNC J6 [R13]	Not taken over: not cabled at SPEAR3 (D4, Table 4)
RF Local Display Panel	Station online/offline and local/remote key switches [R13]	Not taken over; online/offline from EPICS (D6, Table 4)
RF Local Display Panel	Display of PLC-5 states on its LEDs [R13]	Not taken over; the PLC-5 reports them to EPICS (Table 4)
AIM, at I8	The commands to the panel, which the coordinator then sends through the chassis IOC	Command path, §8

From	Function	In the chassis
New	Sum of the permits of the RF MPS, the SPEAR MPS, the orbit interlock, the ARC4, the Waveform Buffer and spare inputs; LLRF9 enable and status; first fault; status to the RF MPS and EPICS [R1], [R4]	§4, §5 and §8
New	RF and HVPS waveforms, comparators and the collector-power trip [R5]	Waveform Buffer, §7

Table 4 lists the legacy signals that the chassis does not take over, with what replaces each, if anything.

Table 4: Legacy signals not taken over.

Signal	Legacy connection	Legacy use	Disposition
Twelve arc-detector fibers	Fast Interlock Chassis receivers; analog signals on the 37-pin cable from its J2 to the AIM's J4	The AIM's arc comparators and latches	The fibers move to the ARC4 at I1 (A1, A2); the AIM's arc channels are disabled before I2 [R14], [R15]
Spare fault	Front LEMO of the Fast Interlock Chassis; pin 34 of the 37-pin cable	The AIM's spare-fault input	Source not identified; moved to an expansion input if still needed, and the AIM's spare fault disabled before I2 (O4)
Test light and trip tests	Fiber transmitters J33 to J44; twelve trip-test lines on J15	Tests of the legacy arc detectors	Not needed; the ARC4 has its own test (A3)
AIM supplies	J15 pins 4 to 6	Power for the Fast Interlock Chassis	Not connected; the chassis has its own supplies (IC-26)
Detector monitor	Front BNC J16 of the Fast Interlock Chassis	Front monitor, not cabled	No monitor output (IC-35)
Display-only lines	X530 J2-06, -09, -10, -11, -19 and -24; the 13 area summaries on J3 other than waveguide pressure and collector; the J3 spares	The panel's LEDs: main focus on, filament, timeout, HV and RF detected, bucking coil on, area summaries	Not connected; the PLC-5 reports these states to EPICS (Table 17)
Local/remote key switch	Panel; X530 J2-12 to PLC-5 input I:030/1 (P12)	Copied to EPICS only (`SRF1:STN:KEY:STAT`)	Dropped (D6): P12 is not connected, so the PLC-5 reads remote
Online/offline key switch	Panel logic; X530 J2-25 to PLC-5 input I:031/7 (P11)	SCR ENABLE; with the station-online request, the PLC-5's online bit	Replaced (D6): P11 is held on, and the station-online command from EPICS acts in the logic as c_{ON} (IC-36)
Beam-abort output	Panel logic, latched until reset; rear BNC J6	The PEP-II beam-abort system [R24]	Dropped (D4): J6 is not cabled at SPEAR3, and the SPEAR MPS takes no status from the RF in hardware [R4]

The interfaces come into use in steps (Table 5). Every end-state input is built in from the start, read and latched, and acts only when its integration has gone in and its act mask is set [R3].

Table 5: Use of the interfaces by integration. A: acts on the outputs; M: read, latched and reported only; —: not connected.

Interface	I2	I3	I4, I5	I6	I7	I8 on
HVPS fibers	A	A	A	A	A, new controller	A
AIM, J15	A	A	A	A	A	—
PLC-5 or RF MPS lines at X530	A, PLC-5	A	A, RF MPS	A	A	A
HVPS voltage and current buffers	A	A	A	A	A	A
Klystron forward-power detector output	A	A	A, RF MPS	A	A	A
ARC4 permit	M, then A	A	A	A	A	A
Waveform Buffer comparators	M, then A	A	A	A	A	A
LLRF9 interlock and HVPS-interlock outputs	—	M	M	A	A	A
LLRF9 enable	—	—	—	A	A	A
SPEAR MPS and orbit permits	—	—	—	A	A	A
RF MPS summary permit and heartbeat	M	M	M	A	A	A
RF MPS reset and status lines	—	—	—	—	—	A
Expansion ports	—	—	—	as needed	as needed	as needed

3 Chassis Architecture

The Interface Chassis and the Waveform Buffer are one chassis with two boards, one per subsystem; the chassis is being raised from 2U to 3U, the prototype is built and the Waveform Buffer board was completed in April 2026 [R1].

Table 6: Architecture requirements.

ID	Requirement	Basis
CH-1	One 19-inch rack chassis, 3U, holds the Interface Chassis board and the Waveform Buffer board. The two boards are functionally independent: a fault or a removed Waveform Buffer board removes only the Waveform Buffer permit.	[R1]
CH-2	No processor, operating system or network lies in any path from a permit input to a permit output. The permit logic is discrete logic or a programmable logic device configured from non-volatile memory at power-up, with no software.	[R1], [R4]
CH-3	The boards exchange only these signals: the Waveform Buffer permit and its comparator states (to the Interface Chassis); the freeze trigger and the buffered HVPS voltage and current (to the Waveform Buffer); all states to the chassis processor for EPICS.	§7

ID	Requirement	Basis
CH-4	One Ethernet interface serves EPICS for both boards (§8). Loss of the network or of the processor changes no permit.	[R1]
CH-5	The chassis stands in B132 where the legacy cables can reach it: the J15 cable from the AIM, the two X530 cables, the three HVPS fibers, the detector cables to the PLC-5 and the ARC4 cables; any cable that cannot reach is replaced by a new run with the same pinout.	[R3]

4 Interface Chassis Inputs and Outputs

Table 7 lists every external signal of the Interface Chassis board. The sections that follow set the requirements of each group; §6 sets the electrical classes named in the last column but one.

Table 7: External signals of the Interface Chassis board.

ID	Signal	Dir.	Partner	Permit or active state	Class	Use
F1	SCR ENABLE	Out	HVPS controller	Lit: triggers enabled	Fiber out	I2 on
F2	KLYSTRON CROWBAR	Out	HVPS controller	Lit: no crowbar	Fiber out	I2 on
F3	STATUS	In	HVPS controller	Lit: controller enabled, no crowbar	Fiber in	I2 on
L1	LLRF9 enable, to Unit 2's interlock input	Out	LLRF9	High: enabled	LLRF9 out	I6 on
L2	LLRF9 status, Unit 1's interlock output at the end of the chain	In	LLRF9	High: not tripped	LLRF9 in	M from I3, A from I6
L3, L4	LLRF9 HVPS-interlock output, Units 1 and 2	In	LLRF9	High: not tripped	LLRF9 in	M from I3, A from I6
M1	RF MPS summary permit	In	PLC-5, then RF MPS	24 V: permit	24 V in	M from I2, A from I6
M2	RF MPS heartbeat	In	PLC-5, then RF MPS	Pulses present	24 V in	M from I2, A from I6
M3	RF MPS reset	In	RF MPS	Rising edge: reset	24 V in	I8 on
M4	Status lines	Out	RF MPS	24 V: permit present or output on	24 V out	I8 on
S1	SPEAR MPS permit	In	SPEAR MPS	24 V: permit	24 V in	I6 on
S2	Orbit interlock permit	In	Orbit interlock	24 V: permit	24 V in	I6 on
A1	Arc permit	In	ARC4 system modules	No arc	Per O3	M at I2, then A

ID	Signal	Dir.	Partner	Permit or active state	Class	Use
A2	Fired-detector identity	In	ARC4 system modules	—	Per O3	I2 on
A3	Test, reset	Out	ARC4 test and system modules	—	Per O3	I2 on
X1, X2	Expansion, TTL	In	As assigned	High: permit	TTL in	As needed
X3, X4	Expansion, 24 V	In	As assigned	24 V: permit	24 V in	As needed
V1–V6	Six AIM commands	In	AIM, J15	Table 15	TTL in	I2 to I7
V7, V8	AB summary and heartbeat	Out	AIM, J15	Table 15	TTL out	I2 to I7
P1–P12	PLC-5 lines at X530	In, out	PLC-5, then RF MPS	Table 17	24 V in, out	I2 on
N1, N2	HVPS voltage and current monitors	In	HVPS monitor board	Analog	Analog in	I2 on
N3, N4	Buffered HVPS voltage and current	Out	PLC-5, then RF MPS	Analog	Analog out	I2 on
N5	Klystron forward RF, on the cable now at J32	In	Klystron forward coupler	RF	RF in	I2 on
N6	Klystron forward-power level, on the cable now at J31	Out	PLC-5, then RF MPS	Analog	Analog out	I2 on
E1	Ethernet	In, out	Controls network	—	—	I2 on

4.1 HVPS Controller Fibers

The three fibers run between the chassis in B132 and the trigger interconnect boards of the active HVPS controller in B118, through a switchbox that selects the unit [R12], [R17]. SCR ENABLE goes to the Right Side Trigger Interconnect board, which passes it to the Left Side board; while it is dark both boards inhibit the triggers of phases A and C, and phases B^{+} and B^{-} keep firing so that the filter inductors discharge. A dark KLYSTRON CROWBAR fiber starts a 2.37 s pulse on the Left Side board that fires the crowbar and inhibits the triggers. STATUS is lit while the controller's 24 V fault/enable output (PLC slot 5, output 0) is high and no crowbar trigger is present; it says nothing about whether the thyristors are firing [R10], [R11]. The new HVPS controller of I7 keeps the three fibers and their conventions [R3].

Table 8: HVPS fiber requirements.

ID	Requirement	Basis
IC-1	Two fiber transmitters (SCR ENABLE, KLYSTRON CROWBAR) and one receiver (STATUS), with ST connectors, compatible with the existing fibers through the switchbox and with the receivers and transmitter of both the legacy and the new HVPS controller: Broadcom HFBR-1414 transmitters, the part that the panel uses, and an HFBR-2412 receiver, or equivalents of the same family. Each transmitter gives at least the optical power of the panel's transmitter today (IC-3).	[R8], [R12], [R13]
IC-2	Light means permit or no fault, as today: SCR ENABLE lit enables the triggers, KLYSTRON CROWBAR lit means no crowbar, STATUS lit means the controller can run. Both transmitters are dark when the chassis is unpowered.	[R10], [R11]
IC-3	The optical power at each receiver of the link is measured at installation and has at least 3 dB margin over the receiver's sensitivity (*proposed*).	—
IC-4	The STATUS receiver is a latched permit input (F3 in the logic). Its state and latch are reported. The logic relies on STATUS being lit whenever the controller is ready to run, whether or not SCR ENABLE is lit, as the legacy controller does; the new controller of I7 keeps this.	[R4], [R10]

4.2 LLRF9 Units

Each LLRF9 unit has an opto-isolated external interlock input, held high to enable operation; a trip from it or from any of 17 internal sources (nine RF overvoltage comparators and eight slow-ADC window comparators) turns off the drive and opens the absorptive switches in the output chains. The interlock output gives $4.75\,\mathrm{V}$ unloaded through $220\,\Omega$ and falls below $0.1\,\mathrm{V}$ when tripped; its return is tied to the LLRF9 chassis. The spare output is set by default as an HVPS interlock, tripped only by slow-ADC events, to protect the klystron. The input level is a factory option: the input is guaranteed high at $2.8\,\mathrm{V}$, $3.9\,\mathrm{V}$ or $17.2\,\mathrm{V}$ for the $3.3\,\mathrm{V}$, $5\,\mathrm{V}$ and $24\,\mathrm{V}$ options, with absolute maxima of $3.8\,\mathrm{V}$, $6.0\,\mathrm{V}$ and $34.5\,\mathrm{V}$; the driver must supply at least $7\,\mathrm{mA}$, and $8\,\mathrm{mA}$ is recommended. The connectors are LEMO EGG.00.302.CLL, pin 1 signal and pin 2 return [R20].

The two units are daisy-chained, as decided in Doc 0 [R1] (D1): the chassis enable drives Unit 2's interlock input, Unit 2's interlock output drives Unit 1's interlock input, and Unit 1's interlock output returns to the chassis as the LLRF9 status. A trip of either unit, or a removed enable, stops the drive of Unit 1 with no chassis logic between the units, and the vendor-native link carries a reflected-power trip of Unit 2 straight to the drive.

Table 9: LLRF9 interface requirements.

ID	Requirement	Basis
IC-5	One isolated enable output (L1) to Unit 2's interlock input, high to enable. The high level lies between the guaranteed-high level and the absolute maximum of the option fitted to Unit 2, at a load current of $8\,\mathrm{mA}$ or more **[O2]**; for the $5\,\mathrm{V}$ option, 4.0–$5.5\,\mathrm{V}$ (*proposed*). The low level is below $0.4\,\mathrm{V}$. The output is current-limited and survives a short circuit.	[R4], [R20]
IC-6	Three isolated status inputs: Unit 1's interlock output at the end of the chain (L2) and the HVPS-interlock outputs of both units (L3, L4), with a threshold between $1.0\,\mathrm{V}$ and $2.5\,\mathrm{V}$ and an input current of at most $2\,\mathrm{mA}$ (*proposed*).	[R20]

ID	Requirement	Basis
IC-7	LEMO connectors that mate with the LLRF9 cables: EGG.00.302.CLL, or EGG.00.303.CLL where a different level must not be cross-connected.	[R20]
IC-8	The chassis does not tap the link from Unit 2 to Unit 1, so that it cannot load that link; which unit tripped first is read from the LLRF9 interlock records in EPICS.	[R1], [R20]

4.3 RF MPS

Until I4 the RF MPS is the PLC-5, from I4 the ControlLogix, which takes over the PLC-5 field wiring and keeps its outputs [R3]. The PLC-5's global interlock sum (output O:031/0, the AB summary) and its heartbeat (O:031/5) reach the chassis on the X530 lines J3-09 and J3-21 and serve as the RF MPS summary permit and heartbeat. The heartbeat is a pulse one scan long each time a 50 ms timer expires; it stops when an analog or thermocouple module reports a read error [R19].

Table 10: RF MPS interface requirements.

ID	Requirement	Basis
IC-9	The RF MPS summary permit (M1) is a latched permit input taken from the global interlock sum line; it acts from I6 (D2).	[R1], [R4], [R19]
IC-10	A heartbeat watchdog (M2) accepts pulses of 1 ms or longer and removes its permit when no rising edge arrives within the timeout. The timeout is set by a component or jumper between 0.1 s and 1 s, to three to five times the measured period [O6]; the measured period is reported. The watchdog acts from I6 (D2).	[R1], [R19]
IC-11	An RF MPS reset input (M3) resets the latches on its rising edge (§5.3).	[R4]
IC-12	Status lines to the RF MPS (M4), one per item of Table 30, 24 V sourcing outputs (§6), 24 V meaning permit present or output on.	[R1], [R4]

4.4 SPEAR MPS and Orbit Interlock

Both systems provide a 24 V permit that today reaches only VXI Slot 5, where losing either asserts RF_FAULT and the RFP removes the drive [R2], [R4]. With the LLRF9 driving, the chassis must remove the LLRF9 enable and SCR ENABLE when either permit drops [R4].

Table 11: External permit requirements.

ID	Requirement	Basis
IC-13	Two isolated 24 V permit inputs (S1, S2), latched. Their input current is the lowest that keeps the noise margin of §6, so that they can be paralleled with the Slot 5 inputs until I8; the source budget is checked [O7].	[R4]
IC-14	They act from I6 and remove the LLRF9 enable and SCR ENABLE.	[R3], [R4]

4.5 ARC4

The ARC4 detects the light of an arc at the four cavity windows (two fibers each), the klystron output window and the circulator. Its two system modules present a fail-safe, active-high arc permit and the identity of the

detector that fired; its test module lets the chain be exercised without an arc. Only the system modules and the test module connect to the chassis [R1].

Table 12: ARC4 interface requirements.

ID	Requirement	Basis
IC-15	The arc permit (A1) is a latched permit input of the class that the ARC4 output requires: an isolated 24 V input for an active output, or a wetted input (the chassis supplies an isolated 24 V through the contact) for a dry contact or solid-state relay [O3].	[R1]
IC-16	The fired-detector identity (A2) is read for ten channels, as parallel bits or an encoded word, latched with the arc permit and reported in the first-fault record and to EPICS.	[R1]
IC-17	Test and reset lines to the ARC4 (A3) are driven from EPICS commands through the chassis processor; a chassis reset also resets the ARC4 latch.	[R1]
IC-18	The arc permit acts after a logged run without nuisance trips, by its act mask.	[R3]

4.6 Beam Abort

The panel forms a beam abort from the VXI beam abort (V4), the HVPS STATUS fiber and the two coil summaries, latches it until reset and drives it through a 49.9 Ω resistor to rear BNC J6 [R13]. The output was made for the PEP-II beam-abort system [R24]; at SPEAR3 J6 is not cabled. The SPEAR MPS and the orbit interlock dump the beam by removing the RF permit (S1, S2) and take no status from the RF in hardware [R4], so the chassis has no beam-abort output (D4).

Table 13: Beam-abort requirements.

ID	Requirement	Basis
IC-19	No beam-abort output (D4).	[R4], [R24]
IC-20	The VXI beam abort V4 is received until I7, shown on the front panel and reported to EPICS; it acts on no output.	[R13]

4.7 Expansion Ports

Table 14: Expansion port requirements.

ID	Requirement	Basis
IC-21	Two TTL permit inputs (X1, X2) on BNC connectors, input impedance about 1 kΩ, so that a fast source can be terminated in 50 Ω with a terminator at the chassis.	[R4]
IC-22	Two 24 V permit inputs (X3, X4).	[R4]
IC-23	All four are latched permit inputs with act masks, off until assigned.	[R4]

4.8 AIM Cable J15 (I2 to I7)

The AIM drives 23 of the 25 lines of J15: the six commands; its fused $\pm 12\,\mathrm{V}$ and $5\,\mathrm{V}$, which power the whole Fast Interlock Chassis; and twelve arc trip-test lines. Pins 10 and 23 return the AB summary and the heartbeat to it, and pins 17 and 18 are grounds. The Fast Interlock Chassis loads each command with a $2\,\mathrm{k}\Omega$ pull-up and a Schmitt trigger and sends it to the panel as a fiber that is lit while the line is low: for a request, or, on pin 15, for no abort (the AIM's net there is ABORT_BAR, inverted by its output buffer). It drives pins 10 and 23 from open-collector fiber receivers with $1\,\mathrm{k}\Omega$ pull-ups, low while the fiber is lit, into the AIM's Schmitt-trigger inputs with $10\,\mathrm{k}\Omega$ to ground [R12], [R14], [R15]. Table 15 gives the resulting sense of each line. The AIM latches an AB summary fault, which trips the legacy station and blocks its turn-on, and an AB heartbeat fault, which raises a major alarm [R23]; V7 and V8 are therefore needed for as long as the AIM runs the legacy station.

Table 15: J15 lines [R12], [R14], [R15].

ID	J15 pin	Signal; panel request	Active on J15	Use in the chassis
V1	1	FILAMENT_ON; filament and focus request	Low: request	To the filament-request line (P8)
V2	14	SOLENOID_ON; station online request	Low: request	To the solenoid-request line (P7); command input of the logic (c_{ON})
V3	2	HVPS_ON; HVPS on request, a level held by the AIM	Low: request	Command input of the logic (c_{HV})
V4	15	BEAM_ABORT; VXI beam abort	High: abort	Shown and reported only (IC-20)
V5	3	FILAMENT_TIMEOUT; filament timeout bypass, a pulse	Low: request	To the timeout-request line (P9)
V6	16	FAULT_RESET; reset, a pulse	Low: reset	To the reset-request line (P10); resets the chassis latches
V7	10	AB_SUMMARY	Low: no fault	From the AB summary line (P5)
V8	23	HEART_BEAT	Low during each pulse	From the heartbeat line (P6)
—	4, 5, 6	The AIM's fused $\pm 12\,\mathrm{V}$ and $5\,\mathrm{V}$	—	Not connected
—	17, 18	AIM grounds	—	Reference of the J15 circuits
—	7–9, 11–13, 19–22, 24, 25	Arc trip-test lines TT0 to TT11	Low pulses	Not used

Table 16: AIM interface requirements.

ID	Requirement	Basis
IC-24	A 25-pin D connector that takes the existing J15 cable unchanged, with the pinout of the Fast Interlock Chassis.	[R12], [R14]

ID	Requirement	Basis
IC-25	The six command inputs present the load of the Fast Interlock Chassis: a 2 kΩ pull-up to 5 V on the AIM side of the isolation and a Schmitt-trigger input, so that an open line reads as no request and an open pin 15 as abort. They pass the AIM's shortest pulse without loss [O4]. Pins 10 and 23 are driven as the Fast Interlock Chassis drives them, open-collector with 1 kΩ pull-ups to 5 V: low for no fault and during each heartbeat pulse, into the AIM's 10 kΩ inputs and, on pin 10, its 1 μF.	[R12], [R14], [R15]
IC-26	Pins 4 to 6, the AIM's fused supplies, are not connected, and the chassis draws no power from the AIM; pins 17 and 18 are the reference of the J15 circuits; the twelve trip-test lines are not used.	[R14], [R15]
IC-27	A command path from the chassis IOC takes over the six commands when the AIM is retired at I8 (§8.3); a jumper selects J15 or the IOC, and its state is reported.	[R3]

The Fast Interlock Chassis has other connections that the chassis does not take over: the legacy arc-detection front end, whose tunnel fibers move to the ARC4 at I1 and whose 37-pin cable goes to the AIM's J4; twelve test-light fibers; the front monitor output J16 of the detector, which is not cabled; and a spare-fault input on a front LEMO connector, which reaches the AIM through the same 37-pin cable and is cabled to a source not yet identified [R14], [R15]. Without the Fast Interlock Chassis the AIM reads its open J4 as a spare fault. The AIM's arc channels and spare fault are therefore disabled before I2, and the source of the spare-fault input is moved to an expansion input if it is still needed (O4).

4.9 PLC-5 Lines at X530

The panel's two 25-pin connectors J2 and J3 are cabled to cross-connect X530 in rack B132-14; the lines are 24 V and run to the PLC-5's 1771-IBN input module (10–30 V DC, sinking) and 1771-OBN output module (10–30 V DC, sourcing) [R13], [R16], [R18], [R21]. Table 17 gives each line by the panel schematic's pin, the PLC-5 point from the wiring diagrams and its use in the program [R18], [R19], and what the chassis does with it. The chassis takes only the lines that the panel's logic, fiber links and analog buffers use, and holds the online-mode line on (IC-36). The local-mode line and the lines that the panel routes only to its front-panel LEDs are left unconnected; the PLC-5 reports those states to EPICS itself [R13], [R19]. The drawings agree on the signal names but not on the pin and terminal numbers [O5]. From I4 the RF MPS keeps these lines on the same field wiring.

Table 17: Lines at X530 (pins of SD-340-311-01).

ID	Line (pin)	PLC-5 point	Program use	Chassis
P1	Main-focus summary (J2-07, "Fault15")	O:031/12 (output 032)	Main-focus summary	Latched permit input, crowbar group (p_{FM})
P2	Bucking-coil summary (J2-20, "Fault15")	O:031/13 (output 033)	Bucking-coil summary	Latched permit input, crowbar group (p_{BC})
P3	Waveguide pressure (J3-06)	O:030/12	Waveguide air-pressure summary	Latched permit input (p_{WGA})
P4	Collector (J3-16)	O:031/6	No collector-power trip	Latched permit input (p_{COL})

ID	Line (pin)	PLC-5 point	Program use	Chassis
P5	AB summary (J3-09)	O:031/0	Global interlock sum	RF MPS summary permit M1; driven to the AIM (V7)
P6	AB heartbeat (J3-21)	O:031/5	One scan high after each 50 ms timer	Heartbeat M2; driven to the AIM (V8)
P7	Solenoid request (J2-18)	I:031/3	With I:031/7, the online bit	Output from command V2
P8	Filament request (J2-21)	I:031/1	Starts the warm-up; enables the focus outputs	Output from command V1
P9	Timeout request (J2-22)	I:031/2	Timer bypass, held while the filament is on	Output from command V5
P10	Reset request (J2-23)	I:030/0	Clears the fault latches	Output from command V6
P11	Online mode (J2-25)	I:031/7	With I:031/3, the online bit	Output, held on while the chassis is powered (IC-36)
P12	Local mode (J2-12)	I:030/1	Copied to EPICS only	Not connected; the PLC-5 reads remote (IC-36)
N3	HVPS voltage out (J2-03; return J2-16)	AI 2, ch. 13	Collector power; HV detected	Buffer output
N4	HVPS current out (J2-04; return J2-17)	AI 2, ch. 14	Collector power	Buffer output
—	Grounds (J2-05, -08, -13; J3-12, -13)	—	—	Connected as on the panel
—	Display only: main focus on (J2-06), filament (J2-09), timeout (J2-10), HV detected (J2-11), bucking coil on (J2-19), RF detected (J2-24); the 13 area summaries on J3 other than waveguide pressure and collector; the spares (J3-10, -11, -22 to -25)	O:031/1, /2, /3, /7, /10; O:030/0 to /11, /13 to /16	Shown on the panel only	Not connected

The pairing follows the signal names. The Fast Interlock Chassis sends FILAMENT_ON and SOLENOID_ON on separate fibers [R14]; the panel drives the filament-request line J2-21 from receiver U14 (“filament on”) and the solenoid-request line J2-18 from receiver U13 (“solenoid on”) [R13]. V1 therefore reaches the filament request and V2, with the online-mode line P11, the PLC-5’s online bit, which is also how the legacy software treats the two commands [R22]. The fiber-cable diagram, and J. Sebek’s notes that follow it, label the Fast Interlock Chassis ends of the U13 and U14 fibers the other way round [R12], [R17]; those two labels are swapped, and the copy of the drawing in the repository carries a note. The baseline test of the panel confirms the pairing (Doc M I0-T1).

The panel’s HVPS voltage and current inputs (J2-01 and J2-14, J2-02 and J2-15) come from the HVPS controller and are covered in §4.10. The panel schematic numbers J3 in the reverse pin order of the cross-connect diagram, and the cross-connect diagram puts a “Waveguide Air MPS” line on the schematic’s spare 1 (J3-22) [R11], [R13], [R16].

Table 18: PLC-5 interface requirements.

ID	Requirement	Basis
IC-28	Two 25-pin D connectors that take the existing J2 and J3 cables unchanged, with the pinout of SD-340-311-01.	[R13]
IC-29	Every line of Table 17 that the chassis takes keeps its direction, level and sense: a line the PLC-5 drives is an isolated 24 V input of the chassis; a line the panel drove is an isolated 24 V sourcing output, able to drive a 1771-IBN input and later the ControlLogix input that replaces it. The display-only lines are left unconnected.	[R13], [R18]
IC-30	Each command from the AIM reaches its PLC-5 line (V1, V2, V5 and V6 to P8, P7, P9 and P10), as on the panel; the online-mode line P11 is held on and the local-mode line P12 is not connected (IC-36); the AB summary and heartbeat lines are passed on to the AIM (V7, V8).	[R13]
IC-31	Every line that the chassis takes is shown on its front panel and reported to EPICS (§8).	[R13]

4.10 Analog Signals

The HVPS monitor board in the HVPS controller provides the voltage at $10\,\mathrm{kV/V}$ and the current at $5\,\mathrm{A/V}$ [R2]; the panel buffers both into the PLC-5's analog module, whose program takes $0.024\,331\,\mathrm{kV}$ and $0.012\,4\,\mathrm{A}$ per count, the same scale for a $\pm 10\,\mathrm{V}$ input range, and calls the high voltage detected above 400 counts (about $9.7\,\mathrm{kV}$) [R19]. The program also takes the klystron forward power, on analog input 2, channel 12 (RF PWR LEVEL), from the rear output J31 of the Fast Interlock Chassis detector [R18], as $(0.017\,c)^2$ kW for c counts, a detector output proportional to the RF amplitude, about $0.144\,\mathrm{V}$ per $\sqrt{\mathrm{kW}}$ on that range; the RF power is detected above 175 counts (about $9\,\mathrm{kW}$) [R19]. In that chassis the RF from the klystron forward coupler enters at the rear on J32; a detector module gives 0 to $-10\,\mathrm{V}$ and an inverting amplifier 0 to $10\,\mathrm{V}$, through $1\,\mathrm{k\Omega}$ to J31 and, through a follower and another $1\,\mathrm{k\Omega}$, to the front BNC J16, which is not cabled. The detector and the amplifier run from the AIM's supplies on J15 [R14].

Table 19: Analog signal requirements.

ID	Requirement	Basis
IC-32	Differential inputs for the HVPS voltage and current monitors (N1, N2) on the panel's pins, input range $\pm 10\,\mathrm{V}$, common-mode range at least $\pm 10\,\mathrm{V}$ and common-mode rejection at least $60\,\mathrm{dB}$ at $60\,\mathrm{Hz}$ (*proposed*).	[R13]
IC-33	Buffered single-ended outputs (N3, N4) to the PLC-5 at unity gain, gain error at most $0.1\,\%$ and offset at most $5\,\mathrm{mV}$ (*proposed*), so that the scales of 10 kV and 5 A per volt reach the PLC-5 unchanged; bandwidth at least $1\,\mathrm{kHz}$.	[R2], [R19]
IC-34	Copies of both signals to the Waveform Buffer through separate buffers, with a bandwidth of at least $50\,\mathrm{kHz}$ (*proposed*); a fault on one copy does not disturb the other.	[R3], [R5]

ID	Requirement	Basis
IC-35	A klystron forward-power detector in place of the Fast Interlock Chassis's: RF input (N5) for the cable now on J32; output (N6), 0 to 10 V through 1 kΩ, on a connector that takes the PLC-5 cable now on J31 (pin A signal, pin B ground); the transfer curve of the legacy detector, about $0.144\,\mathrm{V}$ per $\sqrt{\mathrm{kW}}$ on the PLC-5's scale, within 2 % of reading over the operating range (*proposed*), taken from the spare Fast Interlock Chassis or from archived readings before design (O4). The chassis powers it; no front monitor output is needed. It serves the PLC-5 and then the RF MPS for as long as they compute the collector power from it.	[R12], [R14], [R18], [R19]

4.11 Operator Controls and Display

Table 20: Operator controls and display.

ID	Requirement	Basis
IC-36	No key switches (D6). The online-mode line P11 is held on while the chassis is powered, so that the PLC-5's online bit follows the station-online command V2 alone, which the operator sets from EPICS (`SRF1:STN:AIM:SOLENOID`) until I8; the same command enters the logic as c_{ON}. If the chassis loses power, P11 drops and the legacy software takes the station offline. The local-mode line P12 is not connected, so the PLC-5 reads remote.	[R19], [R22], [R23]
IC-37	Front-panel indicators for every permit input (present, latched), every output, the first fault, the heartbeat, each act mask that is off and the six AIM commands. The panel's display of PLC-5 states is not reproduced.	[R1], [R13]
IC-38	No front-panel reset button (D6). Every reset comes from FAULT_RESET on J15 (I2 to I7), the RF MPS reset M3 or an EPICS command (LG-2); bench tests reset through the IOC or the simulators.	—

5 Interface Chassis Logic

The panel's GAL22V10 forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort from the HVPS-on request, the VXI and HVPS beam aborts, the station online key switch, the main-focus and bucking-coil summaries and the waveguide air-pressure and collector-power summaries, and two flip-flops latch the beam abort and the crowbar until reset [R13]; its equations are not documented. The equations below are written from our understanding of the system's required behavior: J. Sebek's specification and notes [R4], [R8]–[R11], the HVPS controller's response to the fibers [R10], the PLC-5 program [R19] and the protection layers of Doc 0 [R1]. They are reviewed and compared with the recorded behavior of the panel before the logic is frozen **[O1]**.

5.1 Notation

Every symbol in the equations is binary: 1 means a permit present, a request active or an output on; $\wedge$ is AND, $\vee$ OR and $\neg$ NOT. Each permit input k has three states: $x_k = 1$ while its permit is present (24 V on the line, the fiber lit or the LLRF9 output high), $a_k = 1$ when its act mask is set (LG-8), and $L_k = 1$ while its fault latch is set. The latch is set when $x_k = 0$ and is cleared by a reset only if $x_k = 1$ at that moment.

The effective permit of the input is

$$p_k = \neg\,(a_k \wedge L_k), \tag{1}$$

so an input whose mask is off never removes a permit, while its latch still records every fault. The three LLRF9 inputs are latched inside the LLRF9 itself, so in the permit path they act without a chassis latch,

$$q_k = \neg a_k \vee x_k,$$

and have record latches only (LG-6). Command inputs request an action and do not latch. Table 21 lists every symbol with its signal (Table 7) and its source.

Table 21: Symbols of the logic.

Symbol	Equal to 1 when	Signal	Source
p_{MPS}	The RF MPS gives its summary permit, the global interlock sum (AB summary)	M1 (P5, J3-09)	PLC-5 O:031/0, then the RF MPS
p_{HB}	Heartbeat pulses arrive within the watchdog timeout (IC-10)	M2 (P6, J3-21)	PLC-5 O:031/5, then the RF MPS
p_{SMPS}	The SPEAR MPS gives its beam permit	S1	SPEAR MPS
p_{ORB}	The orbit interlock gives its permit	S2	Orbit interlock
p_{ARC}	No arc	A1	ARC4 system modules
p_{WB}	No Waveform Buffer comparator or collector-power trip (WB-7)	Between the boards	Waveform Buffer
p_{X1} to p_{X4}	Each assigned expansion permit present	X1 to X4	As assigned
P_S	All ten permits above: the station permit, (2)	—	—
p_{ST}	STATUS lit: the controller enabled by its PLC and no crowbar trigger; the latched effective permit	F3	HVPS controller
p_{FM}	Main-focus summary on: filament requested, main-focus voltage and current inside their limits, supply's internal contact closed	P1 (J2-07)	PLC-5 output 032 (O:031/12)
p_{BC}	Bucking-coil summary on: filament requested, bucking-coil voltage and current inside their limits	P2 (J2-20)	PLC-5 output 033 (O:031/13)
P_{CB}	Every further input assigned to the crowbar group by jumper gives its permit; 1 when none is assigned, the default (D3)	Jumpers (LG-8)	—
p_{WGA}	Waveguide air-pressure summary on	P3 (J3-06)	PLC-5 O:030/12
p_{COL}	No collector-power trip: the PLC-5 latches one when the HVPS power less the klystron output power stays above 1 200 kW for 0.7 s	P4 (J3-16)	PLC-5 O:031/6
q_u	LLRF9 status high: Unit 1's interlock output, high only while neither unit is tripped and the enable is present	L2	LLRF9 Unit 1
q_{h1}, q_{h2}	The HVPS-interlock outputs of Units 1 and 2 high	L3, L4	LLRF9

Symbol	Equal to 1 when	Signal	Source
c_{HV}	HVPS on requested	V3 (J15 pin 2, HVPS_ON) until I7; the chassis IOC from I8	State machine through the AIM, then the coordinator
c_{ON}	Station online requested	V2 (J15 pin 14, SOLENOID_ON) until I7; the chassis IOC from I8	Operator through EPICS and the AIM, then the coordinator
EN	LLRF9 enable on	L1	To Unit 2's interlock input
CB	KLYSTRON CROWBAR lit: no crowbar	F2	To the HVPS controller
SCR	SCR ENABLE lit: triggers enabled	F1	To the HVPS controller

5.2 Equations

The station permit is the product of every permit that must stop the RF:

$$P_S = p_{MPS} \wedge p_{HB} \wedge p_{SMPS} \wedge p_{ORB} \wedge p_{ARC} \wedge p_{WB} \wedge p_{X1} \wedge p_{X2} \wedge p_{X3} \wedge p_{X4}. \tag{2}$$

The outputs are:

$$EN = P_S \wedge p_{ST}, \tag{3}$$

$$CB = p_{FM} \wedge p_{BC} \wedge P_{CB}, \tag{4}$$

$$SCR = c_{HV} \wedge c_{ON} \wedge CB \wedge p_{WGA} \wedge p_{COL} \wedge P_S \wedge q_u \wedge q_{h1} \wedge q_{h2}. \tag{5}$$

In words:

- The LLRF9 enable needs the station permit and the HVPS STATUS; it reaches Unit 1 through Unit 2 (§4.2).
- KLYSTRON CROWBAR stays lit while both coil summaries, and any input added to the crowbar group, give their permits. They are latched, so a coil fault keeps it dark until a reset.
- SCR ENABLE needs the HVPS-on request, the station-online command, a lit KLYSTRON CROWBAR, the waveguide air-pressure and collector-power summaries, the station permit and, from I6, the LLRF9 status and HVPS-interlock outputs.

An input whose act mask is off drops out of every equation ((1) and q_k); Table 5 gives when each input acts.

Table 22: Reasons for each term.

Output	Term and reason	Basis
LLRF9 enable	Every station permit removes the drive: the RF MPS, the SPEAR MPS, the orbit interlock, an arc, a Waveform Buffer trip, the heartbeat and the HVPS STATUS, which goes dark when the crowbar fires	[R4], [R8]

Output	Term and reason	Basis
LLRF9 enable	No dependence on SCR ENABLE or on the LLRF9 status, so that the units can be reset while the HVPS is off and no permit waits on itself. A trip of Unit 2, such as on reflected power, reaches the drive of Unit 1 through the daisy chain, not through the chassis (D1)	[R1], [R4], [R20]
KLYSTRON CROWBAR	Fired, and latched, on a main-focus or bucking-coil summary fault: a coil current outside its limits can steer the beam into the tube body, as the panel does today. Kept lit otherwise, as J. Sebek recommends	[R9], [R11], [R12], [R13]
SCR ENABLE	The HVPS-on request and the waveguide air-pressure and collector-power summaries, as in the panel; the station-online command in place of the panel's online key, so that taking the station offline from EPICS removes SCR ENABLE at once (D6)	[R11], [R13]
SCR ENABLE	Dark whenever KLYSTRON CROWBAR is dark, as a redundant inhibit	[R9]
SCR ENABLE	Every station permit, and from I6 the LLRF9 status and HVPS-interlock outputs: with the drive off the collector would take the full beam power	[R4], [R8], [R11]
SCR ENABLE	Not the HVPS STATUS: a dark STATUS means that the controller has already inhibited its triggers or fired the crowbar, and from I6 STATUS removes the LLRF9 enable and through it SCR ENABLE	[R4], [R8], [R10]

Two consequences are expected and compared with the baseline of the panel (Doc M I0-T1) [R3]: the VXI beam abort does not remove SCR ENABLE, because the state machine forces it in TUNE, which runs with high voltage [R3]; and with the filament off the coil summaries are off, so KLYSTRON CROWBAR stays dark and latched until the reset that comes with the next turn-on, as in the panel if its flip-flop latches the same way.

5.3 Latching, Reset and First Fault

Table 23: Latching, reset and first-fault requirements.

ID	Requirement	Basis
LG-1	Every permit input has a fault latch, set by the input whatever its act mask, and cleared only by a reset while the input is healthy.	[R4]
LG-2	A reset acts on its rising edge and resets all latches at once. It comes from FAULT_RESET on J15 (I2 to I7), the RF MPS reset (M3) or an EPICS command; there is no front-panel reset (IC-38). A reset line held active does not hold the latches clear; a reset held for more than 10 s is reported (*proposed*).	[R1], [R4]
LG-3	The crowbar latch is cleared by the same reset, which also resets the ARC4 latch and the Waveform Buffer comparators.	[R1], [R13]

ID	Requirement	Basis
LG-4	A first-fault register captures the latches of all acting inputs at the first fault after a reset and holds them until the next reset. Faults closer together than 100 ns are recorded together (*proposed*).	[R1], [R4]
LG-5	A second register does the same for inputs whose act mask is off, so that a monitor-only run records what would have tripped.	[R3]
LG-6	The LLRF9 inputs have record latches and enter both registers, although they act directly in the permit path.	[R4]
LG-7	A fault that a removed output causes, such as an LLRF9 trip after its enable was removed, does not displace the first fault.	[R4]

5.4 Act Masks

Table 24: Act mask requirements.

ID	Requirement	Basis
LG-8	One act mask per permit input, the crowbar-group selection of each input and the J15 or IOC command source are set by jumpers or switches inside the chassis, never by software.	[R3], [R4]
LG-9	Every mask is read back, shown on the front panel when off and reported to EPICS, so that the configuration of each integration is recorded.	[R3]
LG-10	The masks follow Table 5; the configuration of each integration is part of its entry criteria.	[R3]

5.5 Recovery Sequence

The LLRF9 trips whenever the chassis removes its enable, so the chassis must let the station restart once the HVPS is safe [R4]. With (3) the enable returns as soon as the station permits and STATUS are healthy and reset, whatever the state of SCR ENABLE and of the LLRF9 units. This relies on STATUS being lit whenever the HVPS controller is ready to run, whether or not SCR ENABLE is lit, as with the legacy controller [R10] (IC-4): a STATUS that waited for SCR ENABLE would never let the enable return. A restart after a fault is: the fault cleared and the HVPS controller ready, with STATUS lit; a chassis reset; the LLRF9 interlock reset from EPICS, Unit 2 before Unit 1 because Unit 1's input waits on Unit 2's output; then the HVPS-on request lights SCR ENABLE. Until I8 the legacy state machine issues the reset through FAULT_RESET; afterwards the coordinator does.

6 Electrical, Timing and Fail-Safe Requirements

Table 25: Electrical classes.

Class	Requirement	Basis
24 V in	Isolated by an optocoupler such as the Broadcom ACSL-6xx0 family (about 50 ns), with a series resistor sized for 7–10 mA at 24 V (2.2 kΩ to 2.8 kΩ) unless a lower current is required (IC-13). On above 15 V, off below 5 V; off with up to 1 mA leakage from a PLC output; withstands 35 V and reverse polarity (*proposed* levels).	[R4], [R7], [R8]
TTL in	Isolated Schmitt-trigger input, threshold 1.2–2.0 V, withstands ±15 V (*proposed*). AIM inputs (V1 to V6): a 2 kΩ pull-up to 5 V, as on the Fast Interlock Chassis (IC-25); expansion inputs (X1, X2): about 1 kΩ to ground (IC-21).	[R4], [R14]
LLRF9 in	As IC-6.	[R20]
Fiber in, out	As IC-1 and IC-2.	[R8]
24 V out	Isolated sourcing output, at least 20 V at 20 mA, leakage at most 0.5 mA when off, short-circuit protected (*proposed*); able to drive the 1771-IBN and its ControlLogix replacement.	[R18]
LLRF9 out	As IC-5.	[R20]
TTL out	Isolated open-collector output with a 1 kΩ pull-up to 5 V, low at most 0.5 V while sinking 8 mA (*proposed*); drives the AIM's 10 kΩ inputs, and on pin 10 its 1 μF, as the Fast Interlock Chassis does (IC-25).	[R14], [R15]

Table 26: Isolation, timing and fail-safe requirements.

ID	Requirement	Basis
IC-39	Every electrical input and output is isolated from the chassis logic and from every other connector group, by optocoupler, digital isolator or relay; at least 1.5 kV rms (*proposed*). The returns of external signals are not tied to the chassis logic ground.	[R1], [R4]
IC-40	Propagation from a change of any fiber, LLRF9, SPEAR MPS, orbit, ARC4, Waveform Buffer or TTL input to every output it affects: at most 2 μs (*proposed*).	[R1], [R4]
IC-41	Inputs from PLC lines may be filtered for up to 1 ms against contact and switching noise (*proposed*); other inputs reject pulses shorter than 0.2 μs (*proposed*).	—
IC-42	Loss of chassis power, of any internal supply or of the Interface Chassis board removes every permit: both fibers dark, the LLRF9 enable low, 24 V status lines off. A dark KLYSTRON CROWBAR fiber fires the crowbar, as with the panel today.	[R1], [R10]
IC-43	A supply monitor holds every permit off until all internal supplies are in range and the logic is configured, and the latches come up set, so that a reset is needed after power-up.	[R4]
IC-44	Faults of the chassis itself, such as a failed supply or a configuration error, are latched and reported as their own first-fault entry.	—

Table 27 gives the state the chassis takes for each failure of a partner.

Table 27: Fail-safe states.

Failure	Seen by the chassis as	Result
Chassis power lost	—	All permits removed (IC-42)
LLRF9 unit unpowered	LLRF9 status low, through the chain for either unit	SCR ENABLE dark; Unit 1's drive stopped by the chain
HVPS controller off or crowbar fired	STATUS dark	The controller inhibits its own triggers; from I6 the LLRF9 enable removed, latched, and with it SCR ENABLE
RF MPS stops	Heartbeat lost	LLRF9 enable and SCR ENABLE removed, latched (from I6)
Cable to the SPEAR MPS or orbit interlock cut	Permit lost	As above (from I6)
ARC4 unpowered or cable cut	Arc permit lost	As above, once the ARC4 acts
Waveform Buffer board fault	Its permit lost	As above, once it acts
Network or IOC lost	Nothing	No permit changes; status and waveforms unavailable

7 Waveform Buffer

The legacy IQAs digitize 24 detected RF signals at about 1 MSPS into 32-kilosample circular buffers that freeze on a fault; the two LLRF9 units take the 18 most important, and the Waveform Buffer takes the other six and four HVPS signals now watched by an oscilloscope in B118, which triggers on a loss of HVPS voltage. A failure of the HVPS has been seen to start about 100 ms before the trip [R5]. Each conditioned signal goes two ways: to a digitizer, and to a comparator whose latched fault removes the permits and freezes the buffers [R5].

Table 28: Waveform Buffer channels [R1], [R5].

Channel	Signal	Conditioning	Comparator	Note
RF 1	Klystron drive power	Detector	High	Proposed; on neither LLRF9 **[O8]**
RF 2	Circulator-load reflected power	Detector	High	
RF 3	Station reference power	Detector	Low	A loss of the reference trips
RF 4	Waveguide load 1 reflected power	Detector	High	
RF 5	Waveguide load 2 reflected power	Detector	High	
RF 6	Waveguide load 3 reflected power	Detector	High	
RF 7	Klystron forward power	Detector	—	Proposed, for the collector power
RF 8	Spare	Detector	High	

Chan-nel	Signal	Conditioning	Comparator	Note
HV 1	HVPS voltage	From IC-34	Overvoltage	$10\,\mathrm{kV/V}$
HV 2	HVPS current	From IC-34	Overcurrent	$5\,\mathrm{A/V}$
HV 3	Inductor 1 voltage	Divider	—	New cable [O9]
HV 4	Inductor 2 voltage	Divider	—	New cable

Table 29: Waveform Buffer requirements.

ID	Requirement	Basis
WB-1	Eight RF inputs on SMA connectors (*proposed*), each with a log detector such as the Mini-Circuits ZX47-40LN+ (10–8 000 MHz, about 60 dB range, 0.5–2.1 V into 1 kΩ, 400 ns rise time) after attenuators chosen from the measured operating power of each signal.	[R5], [R6]
WB-2	Each detector output is amplified to the digitizer range (a gain of about five) with a bandwidth of at least 1 MHz.	[R6], [R7]
WB-3	Four HVPS inputs: the voltage and current from the Interface Chassis buffers (IC-34), and the two inductor voltages through dividers sized to their measured amplitude.	[R3], [R5]
WB-4	Digitizers of at least 12 bits. RF channels at 1 MSPS into circular buffers of at least 32 kilosamples (32 ms); HVPS channels with at least 100 ms of buffer, at 1 MSPS with 128 kilosamples or at 250 kilosamples per second with 32 kilosamples. Channels are sampled within 1 μs of each other (*proposed*).	[R5]
WB-5	All buffers freeze together on the chassis fault summary (the first latch of an acting input of the Interface Chassis), on a Waveform Buffer comparator trip, on an external TTL trigger input and on an EPICS command; the time of the freeze is recorded. At least 75 % of each buffer lies before the trigger (*proposed*). Frozen data are kept until read out and re-armed from EPICS.	[R1], [R5]
WB-6	A latched comparator per channel as in Table 28: high threshold on the RF channels, low threshold on the reference, overvoltage and overcurrent on the HVPS voltage and current. Thresholds are set by potentiometers or remotely and read back; each comparator state is reported.	[R1], [R5]
WB-7	The Waveform Buffer permit to the Interface Chassis is the product of all comparators whose mask is set and of the collector-power trip; it is reset with the chassis latches and can be reset from EPICS.	[R1], [R5]
WB-8	Collector power $P_C = VI - P_F$ from the HVPS voltage and current and the klystron forward power of channel 7, calibrated; a trip when P_C exceeds a limit for longer than a delay, both settable, initially 1 200 kW and 0.7 s as in the PLC-5 program (*proposed*). The computation runs in the board's hardware or programmable logic, not in the IOC. P_C is reported once a second.	[R1], [R5], [R19]
WB-9	The collector-power trip and the comparators act only after a run in which they are logged and P_C is compared with the PLC-5's value.	[R3]
WB-10	Averages of every channel over about one second are reported for the archiver.	[R5]
WB-11	A calibration from detector volts to power is stored for each RF channel, and the scale of each HVPS channel, both applied in the IOC.	[R6]

ID	Requirement	Basis
WB-12	The board as completed in April 2026 is checked against WB-1 to WB-11 at the bench test; any requirement it does not meet is recorded as a deviation.	[R1]

8 Status Reporting and EPICS

8.1 Status Lines to the RF MPS

Table 30: Status lines to the RF MPS, from I8 (*proposed*).

Group	Lines
Effective permits	One per permit input of (2), the HVPS STATUS, the coil summaries, the waveguide air-pressure and collector-power summaries and the three LLRF9 inputs
Outputs	SCR ENABLE, KLYSTRON CROWBAR, LLRF9 enable
First fault	Five-bit code of the first fault and a valid line
Chassis	Chassis healthy (supplies and logic configured)

8.2 Chassis IOC

Table 31: EPICS requirements.

ID	Requirement	Basis
ST-1	One IOC for both boards (Doc M work package SW7), with names that cannot collide with legacy names, such as `SRF1:WFBUF:` for the Waveform Buffer.	[R1], [R3]
ST-2	Reported for every input: raw state, latch, act mask; for every output: state; the two first-fault registers with the time of the first fault to 1 ms; the ARC4 identity; the measured heartbeat period; each reset with its source and time.	[R1], [R4]
ST-3	Waveform Buffer: waveforms with their freeze time, comparator states and thresholds, one-second averages, collector power, arm and re-arm.	[R1], [R5]
ST-4	Commands: reset; re-arm; Waveform Buffer thresholds where set remotely; ARC4 test. No command can set an act mask or create a permit.	[R3], [R4]
ST-5	States are also available from I2, when the RF MPS is still the PLC-5 and has no inputs for the status lines.	[R3]
ST-6	The alarm handler and the archiver take every input, output, latch and the first fault.	[R3]

8.3 Command Path at I8

When the AIM is retired at I8 the coordinator sends the commands through the chassis IOC instead of J15: the HVPS-on request enters the logic as c_{HV}; the station-online request enters it as c_{ON} and, with the filament request, timeout bypass and reset, reaches its RF MPS line [R3]. The VXI beam abort has no

successor (D4). Each command is read back; the commands hold their state if the IOC stops and are off after power-up.

9 Chassis, Power and Mechanics

Table 32: Chassis requirements.

ID	Requirement	Basis
CH-6	Rear panel: the J15 connector; the J2 and J3 connectors; three ST fiber ports; six LEMO connectors for the LLRF9; connectors for the SPEAR MPS and orbit permits, the ARC4, the RF MPS status lines and the expansion ports; the RF input and level output of the klystron forward-power detector, for the cables now on J32 and J31; the eight RF inputs and four HVPS inputs of the Waveform Buffer and its trigger input; Ethernet; power.	This document
CH-7	Front panel: the indicators of IC-37; no key switches and no reset button.	IC-36 to IC-38
CH-8	Every connector labeled with its ID and signal names; a drawing of the as-built chassis and cabling.	[R3]
CH-9	Mains power 120 V AC with internal isolated supplies (*proposed*); the supply monitor of IC-43.	—
CH-10	Cooling for continuous operation in the B132 racks.	—

10 Verification

Table 33: Verification of the requirements.

Requirements	Test	When (Doc M)
Logic, LG-1 to LG-10	Every input combination of (2) to (5) with latching and reset, by simulators of the AIM, PLC-5, LLRF9 and HVPS sides; the first-fault order for faults 100 ns apart; every mask	Bench, I2-T1
IC-1 to IC-38	Every line at its level and sense; optical power; loading of the LLRF9 outputs; the analog gains and the detector curve against the legacy detector	Bench, I2-T1
IC-39 to IC-44	Isolation; propagation time; power loss and input loss driving every output to its safe state; power-up with latches set	Bench, I2-T1
Transition interfaces	Every AIM command through to its PLC-5 line; the AB summary and the heartbeat to the AIM; the coil, waveguide-pressure and collector summaries into the logic; the station online and offline from EPICS through to the PLC-5's online bit and SCR ENABLE, with P11 held on and P12 open; HVPS scaling at the PLC-5 inputs; the detector level at the PLC-5 input	In place, I2-T2

Requirements	Test	When (Doc M)
Equivalence	The responses to an HVPS trip, arcs in the HVPS controller, a collector or waveguide-pressure summary, a VXI beam abort and a coil fault, against the baseline of the panel	With high voltage, I2-T3; baseline I0-T1
WB-1 to WB-12	Channel calibration; buffers and freeze; comparators; collector power against the PLC-5's	Bench and run period, I2-T1, I2-T4
LLRF9 and external permits	Every path of the protection table of Doc M with the LLRF9 driving	I6-T1, I6-T2

A failure-mode analysis of the logic and the outputs is completed before I2, as rule 2 of Doc M requires for any element that generates a permit [R3].

11 Decisions

Table 34: Decisions.

ID	Decision	Proposal	Needed by
D1	LLRF9 interlock topology	Decided (Doc 0): daisy chain, chassis to Unit 2, Unit 2 to Unit 1, Unit 1 back to the chassis (§4.2)	Decided
D2	RF MPS summary permit and heartbeat before I6	Monitor-only, so that the station trips as today, through the AIM and the state machine; acting from I6, when the LLRF9 drives	Production review
D3	Crowbar group	The coil summaries only, as the panel; arcs at the klystron window and the circulator can be added by jumper if the ARC4 gives their identity in hardware. Whether a coil fault with the high voltage off also fires the crowbar follows the baseline of the panel (I0-T1)	Production review
D4	Beam-abort output	Decided (10 November 2026): none (§4.6). The panel's J6, made for the PEP-II beam-abort system, is not cabled at SPEAR3, and the SPEAR MPS takes no status from the RF in hardware	Decided
D5	LLRF9 inputs in the permit path	Direct, with record latches, so that a restart needs one chassis reset (§5.3)	Logic design
D6	Operator controls	Decided (10 November 2026): no key switches and no reset button on the chassis; the station is set online or offline, and reset, from EPICS (IC-36, IC-38). The legacy software needs no change: with the online-mode line held on, the PLC-5's online bit and `SRF1:STN:LOCAL:ON` follow the existing station-online command. Procedures that use the panel's keys change	Decided

A Source Index

Ref	Source	Location
[R1]	Doc 0, SPEAR3 LLRF Upgrade System Design Report, R4	`Designs/tex/doc0-system-design/`
[R2]	Doc L, Legacy System Architecture, R1 (HVPS monitor board SD-730-793-12)	`Designs/tex/docL-legacy-architecture/`
[R3]	Doc M, Subsystem Integration and Migration Plan, R0	`Designs/tex/docM-integration-migration/`
[R4]	J. Sebek, LLRF Interface Chassis specification	`llrf/architecture/llrfInterfaceChassis.docx`
[R5]	J. Sebek, waveform buffers for the LLRF upgrade	`llrf/architecture/WaveformBuffersforLLRFUpgrade.docx`
[R6]	J. Sebek, RF power detector chassis	`llrf/architecture/rfPowerDetector.docx`
[R7]	Analog design components (amplifiers, optocouplers, converters)	`llrf/architecture/analogDesignComponents.docx`
[R8]	J. Sebek, interfaces between the RF system controllers	`hvps/architecture/designNotes/interfacesBetweenRFSystemControllers.docx`
[R9]	J. Sebek, RF system MPS requirements	`hvps/architecture/designNotes/RFSystemMPSRequirements.docx`
[R10]	J. Sebek, fiber-optic connections between the RF system controllers (HVPS controller side)	`hvps/architecture/designNotes/controllerFiberOpticConnections.docx`
[R11]	J. Sebek, fiber-optic connections between the LLRF system and the HVPS controller, Rev 3	`llrf/documentation/fiberOpticCableSignalControlRev3.docx`
[R12]	J. Sebek, LLRF documentation notes, R2 (fibers, panel, Fast Interlock Chassis, J15)	`llrf/documentation/LLRFDocumentationNotesR2.docx`
[R13]	RF Local Display Panel drawings, schematic SD-340-311-01	`llrf/documentation/localPanel/`
[R14]	Fast Interlock Chassis schematics SD-340-308-01 and -02	`llrf/documentation/legacyInterfaceModules/`
[R15]	AIM schematic SD-340-309-01	`llrf/documentation/legacyInterfaceModules/sd3403090102.pdf`
[R16]	Wiring diagram WD-340-330-02 (panel cross-connects)	`llrf/documentation/mpsWiringDiagrams/wd3403300200.pdf`
[R17]	Wiring diagram WD-340-330-29 (panel fibers)	`llrf/documentation/mpsWiringDiagrams/wd3403302900.pdf`
[R18]	Wiring diagrams WD-340-330-21, -22, -23 and -25 (PLC-5 analog inputs 1 and 2, 1771-IBN digital input, 1771-OBN digital output)	`llrf/documentation/mpsWiringDiagrams/`
[R19]	RF MPS PLC-5 program, decoded listings	`llrf/allenBradley/`
[R20]	Dimtel, LLRF9 manual (SPEAR3), interlock and digital I/O	`llrf/llrf9/llrf9_manual_s3_screen.pdf`
[R21]	RF Local Display Panel to cross-connect mapping	`llrf/documentation/LocalPanelToXConnectMapping.xlsx`
[R22]	Legacy state machine and message logger (`rf_states.st`,`v`, `rf_msgs.st`,`v`)	`spear-rf-code-legacy/rfApp/src/seq/`

Ref	Source	Location
[R23]	Legacy station summaries and AIM records (`rf_sumy_stn.db`,v, `rf_digital_All.substitutions`,v, `aim.db`,v)	`spear-rf-code-legacy/rfApp/Db/`
[R24]	PEP-II RF station block diagram BD-340-330-00	`llrf/documentation/legacyArchitecture/bd3403300000.pdf`

B Abbreviations

Abbreviation	Meaning
AB	Allen-Bradley
AIM	Arc Interlock Module (VXI slot 12)
ARC4	AFT microwave arc-detection system
EPICS	Experimental Physics and Industrial Control System
Fast IC	Fast Interlock Chassis 340-308
GAL	Generic array logic, a programmable logic device
HVPS	High-voltage power supply
IOC	EPICS Input/Output Controller
IQA	In-phase, quadrature and amplitude detector module
LLRF	Low-level RF
MPS	Machine protection system
MSPS	Mega-samples per second
PPS	Personnel protection system
RFP	RF Processor module
TTL	Transistor-transistor logic levels
VXI	VME eXtensions for Instrumentation